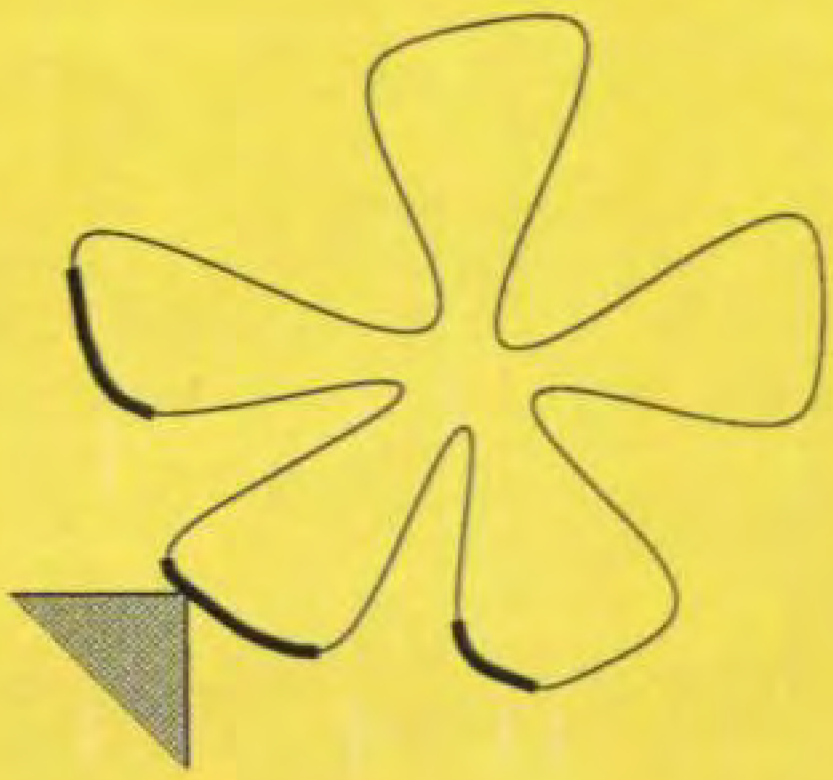
NONLINEAR MULTIOBJECTIVE OPTIMIZATION



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NONLINEAR MULTIOBJECTIVE OPTIMIZATION

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CONTENTS

the origin and the achievements in this field of research from 1776 to 1960 are widely treated in Stadler (1979). A brief summary of the history is also given in Gal and Hanne (1997). There it is demonstrated that multiobjective optimization has its foundations in utility theory and economics, game theory, mathematical research on order relations and vector norms, linear production theory, and nonlinear programming.

Let us mention some further readings. The monographs of Chankong and Haimes (1983b), Cohon (1978), Hwang and Masud (1979), Osyczka (1984), Sawaragi et al. (1985), Steuer (1986) and Yu (1985) provide an extensive overview of the area of multiobjective optimization. Further noteworthy monographs on the topic are those of Rietveld (1980), Vincke (1992) and Zeleny (1974, 1982). A significant part of Vincke (1992) deals, however, with multiattribute decision analysis. The behavioural aspects of multiobjective optimization are mostly treated in Ringuest (1992), whereas the theoretical aspects are extensively handled in the monographs by Jahn (1986a) and Luc (1989).

As far as this book is concerned, the contents are divided into three parts. Part I provides the theoretical background. Chapter 1 leads into the topic and Chapter 2 presents important notation, concepts and definitions in multiobjective optimization with some illustrative figures. Various theoretical aspects appear in Chapter 3. For example, analogous optimality conditions for differentiable and nondifferentiable problems are considered. A solid, conceptual basis and foundation for the remainder of the book is laid. Throughout the book we keep to problems involving only finite-dimensional Euclidean spaces. (Dauer and Stadler (1986) provide a survey on multiobjective optimization in infinite-dimensional spaces.)

The methodology is handled in Part II. Methods are divided into four classes in Chapter 1 according to the role of a (single) decision maker in the solution process. The state of the art in method development is portrayed by describing a number of different methods accompanied by their theoretical background in Chapters 2 to 5. For ease of comparison, all the methods are presented using a uniform notation. The good and the weak properties of the methods are also introduced with references to extensions and applications. The class of interactive methods in Chapter 5 contains most of the methods, and it is the most extensively handled. Linear problems and methods are only occasionally touched on. In addition to describing solution methods, we introduce some implementations. In connection with every method described, some author's comments appear in the concluding remarks. Some of the methods are depicted in more detail and some only mentioned. Appropriate references to the literature are always included.

Part III is Related Issues. After the presentation of a set of different solution methods, some comparison is appropriate in Chapter 1. Naturally, no absolute order of superiority can be given, but some points can be raised. A table comparing some of the features of the interactive methods described is included. In addition, we present brief summaries of some of the comparisons

Part I

TERMINOLOGY AND THEORY

A large variety of solution techniques have been created as to enable the special characteristics of MOLP problems to be taken into account. Here we concentrate on cases where nonlinear functions are included and thus methods for nonlinear problems are needed. Methods and details of MOLP problems are mentioned only incidentally.

Before we define convex multiobjective optimization problems, we briefly write down the definitions of convex functions and convex sets.

Definition 2.1.2. A function $f_i\colon \mathbf{R}^n \to \mathbf{R}$ is *convex* if for all $\mathbf{x}^1, \mathbf{x}^2 \in \mathbf{R}^n$ is valid that $f_i(\beta\mathbf{x}^1 + (1-\beta)\mathbf{x}^2) \leq \beta f_i(\mathbf{x}^1) + (1-\beta)f_i(\mathbf{x}^2)$ for all $0 \leq \beta \leq 1$.

A set $S \subset \mathbf{R}^n$ is *convex* if $\mathbf{x}^1, \mathbf{x}^2 \in S$ implies that $\beta\mathbf{x}^1 + (1-\beta)\mathbf{x}^2 \in S$ for all $0 \leq \beta \leq 1$.

Definition 2.1.3. The multiobjective optimization problem is *convex* if all the objective functions and the feasible region are convex.

A convex multiobjective optimization problem is an important concept in the continuation. We shall also need related generalized concepts, quasiconvex and pseudoconvex functions. The pseudoconvexity of a function calls for differentiability. For completeness, we write down the definitions of differentiable and continuously differentiable functions.

Definition 2.1.4. A function $f_i\colon \mathbf{R}^n \to \mathbf{R}$ is *differentiable* at $\mathbf{x}^*$ if

$$f_i(\mathbf{x}^* + \mathbf{d}) - f_i(\mathbf{x}^*) = \nabla f_i(\mathbf{x}^*)^T\mathbf{d} + \|\mathbf{d}\|\,\varepsilon(\mathbf{x}^*, \mathbf{d}),$$

where $\nabla f_i(\mathbf{x}^*)$ is the *gradient* of f_i at $\mathbf{x}^*$ and $\varepsilon(\mathbf{x}^*, \mathbf{d}) \to 0$ as $\|\mathbf{d}\| \to 0$.

In addition, f_i is *continuously differentiable* at $\mathbf{x}^*$ if all of its *partial derivatives* $\frac{\partial f_i(\mathbf{x}^*)}{\partial x_j}$ $(j = 1, \ldots, n)$, that is, all the components of the gradient are continuous at $\mathbf{x}^*$.

The gradient of f_i at $\mathbf{x}^*$ can also be denoted by $\nabla_x f_i(\mathbf{x}^*)$ to emphasize that the derivation is carried out subject to $\mathbf{x}$.

Now we can define quasiconvex and pseudoconvex functions.

Definition 2.1.5. A function $f_i\colon \mathbf{R}^n \to \mathbf{R}$ is *quasiconvex* if $f_i(\beta\mathbf{x}^1 + (1-\beta)\mathbf{x}^2) \leq \max[f_i(\mathbf{x}^1), f_i(\mathbf{x}^2)]$ for all $0 \leq \beta \leq 1$ and for all $\mathbf{x}^1, \mathbf{x}^2 \in \mathbf{R}^n$.

Let f_i be differentiable at every $\mathbf{x} \in \mathbf{R}^n$. Then it is *pseudoconvex* if for all $\mathbf{x}^1, \mathbf{x}^2 \in \mathbf{R}^n$ such that $\nabla f_i(\mathbf{x}^1)^T(\mathbf{x}^2 - \mathbf{x}^1) \geq 0$, we have $f_i(\mathbf{x}^2) \geq f_i(\mathbf{x}^1)$.

As far as the relations of quasiconvex and pseudoconvex functions are concerned, every pseudoconvex function is also quasiconvex.

The definition of convex functions can be modified for *concave* functions by replacing '$\leq$' by '$\geq$'. Correspondingly, the definition of quasiconvex functions becomes appropriate for *quasiconcave* functions by the exchange of '$\leq$' to '$\geq$' and 'max' to 'min'. In the definition of pseudoconvex functions we replace

Because of the contradiction and possible incommensurability of the objective functions, it is not possible to find a single solution that would be optimal for all the objectives simultaneously. Multiobjective optimization problems are in a sense ill-defined. There is no natural ordering in the objective space because it is only partially ordered (meaning that, for example, $(1,1)^T$ can be said to be less than $(3,3)^T$, but how to compare $(1,3)^T$ and $(3,1)^T$). This is always the case when vectors are compared in real spaces (see also Chankong and Haimes (1983b, pp. 64–67)).

Anyway, some of the objective vectors can be extracted for examination. Such vectors are those where none of the components can be improved without deterioration to at least one of the other components. Edgeworth (1987) presented this definition in 1881. However, the definition is usually called Pareto optimality after the French-Italian economist and sociologist Vilfredo Pareto, who in 1896 developed it further (see Pareto (1964, 1971)). However, in some connections, like in Stadler (1988b), the term *Edgeworth-Pareto optimality* is used for the above-mentioned reason. Koopmans was one of the first to employ in 1951 the concept of Pareto optimality in Koopmans (1971). A more formal definition of *Pareto optimality* is the following:

Definition 2.2.1. A decision vector $\mathbf{x}^* \in S$ is *Pareto optimal* if there does not exist another decision vector $\mathbf{x} \in S$ such that $f_i(\mathbf{x}) \leq f_i(\mathbf{x}^*)$ for all $i = 1, \ldots, k$ and $f_j(\mathbf{x}) < f_j(\mathbf{x}^*)$ for at least one index j.

An objective vector $\mathbf{z}^* \in Z$ is Pareto optimal if there does not exist another objective vector $\mathbf{z} \in Z$ such that $z_i \leq z_i^*$ for all $i = 1, \ldots, k$ and $z_j < z_j^*$ for at least one index j; or equivalently, $\mathbf{z}^*$ is Pareto optimal if the decision vector corresponding to it is Pareto optimal.

In Figure 2.2.1, a feasible region $S \subset \mathbf{R}^3$ and its image, a feasible objective region $Z \subset \mathbf{R}^2$, are illustrated. The fat line contains all the Pareto optimal objective vectors. The vector $\mathbf{z}^*$ is an example of them.

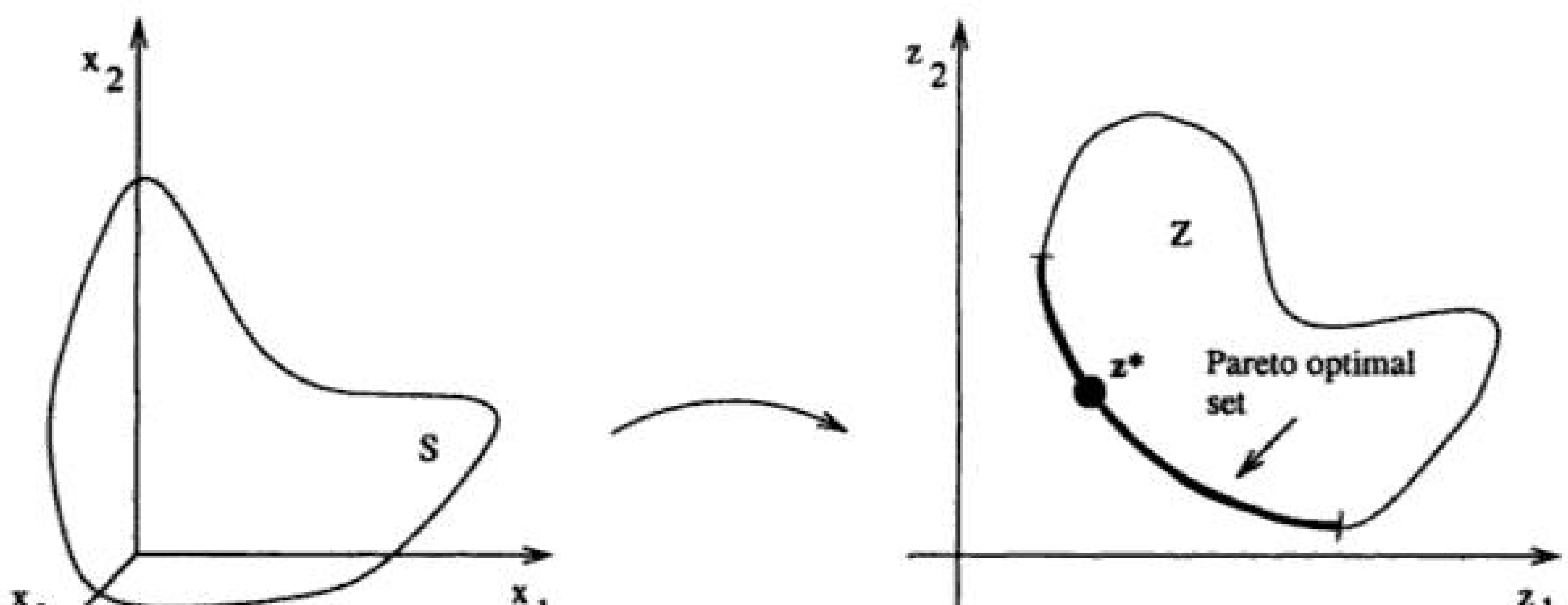

Figure 2.2.1. The sets S and Z and the Pareto optimal objective vectors.

By solving a multiobjective optimization problem we here mean finding a feasible decision vector such that it is Pareto optimal and satisfies the needs and the requirements of the decision maker. Assuming such a solution exists, it is called a *final solution*. However, as stressed in Zionts (1997a, b), it may be difficult for the decision maker to distinguish between good and optimal solutions in real problems. If this is the case, the emphasis should be on finding good solutions (and sometimes, only, on finding solutions).

We do not focus here on the problems of decision making, which is a research area of its own. Interesting topics in this area are, for instance, decision making with incomplete information, validity of the problem formulation and habitual domains. The first of these matter is treated, for example, in Weber (1987). Reasons for incomplete information include lack of knowledge, pressure of time, fear of commitment and matters related to the future.

We usually assume that decision makers are only interested in Pareto optimal points and the rest can be excluded. However, this is not the case if the problem has not been formulated well enough. As already emphasized, non-Pareto optimal solutions may be important if there are some unformulated or hidden objective functions in the mind of the decision maker or some of the objective functions are simply proxies of the objective functions proper (see, for example, Zionts (1997a, b)). In such cases, the Pareto optimal sets of the problem handled and the actual problem which should be solved, do not coincide. Here we assume the mathematical model to be accurate and static so that we can mainly concentrate on Pareto optimal solutions.

A *habitual domain* is defined in Yu (1991) as a set of ways of thinking, judging and responding, as well as the knowledge and experience on which they are based. Yu emphasizes that in order to make effective decisions it is important to expand and enrich the habitual domains of the decision makers. Several ways of carrying this out are presented in Yu (1991, 1995). Understanding, expanding and enriching the domains of thinking is also stressed, for example, in Yu (1994) and Yu and Liu (1997).

2.4. Ranges of the Pareto Optimal Set

Let us for a while investigate the ranges of the set of Pareto optimal solutions. We assume that the objective functions are bounded over the feasible region S.

2.4.1. Ideal Objective Vector

An objective vector minimizing each of the objective functions is called an ideal (or perfect) objective vector.

incorrectly. There may still exist Pareto optimal solutions. However, if, for instance, some component of the ideal objective vector is unbounded and it is replaced by a small but finite number, methods utilizing the ideal objective vector may not be able to overcome the replacement.

Finally, let us look at some examples of the problem of optimizing a function over the Pareto optimal set of a multiobjective optimization problem. This is a more general problem than just looking for the ranges of the Pareto optimal set. In Benson and Sayin (1994), the authors deal with the maximization of a linear function over the Pareto optimal set of an MOLP problem. A general function is minimized over the Pareto optimal set of an MOLP problem in Dauer and Fosnaugh (1995), and a convex function is optimized over the Pareto optimal set of linear objective functions and a convex feasible region by duality techniques in Thach et al. (1996). Maximization of a function over the Pareto optimal set is also considered in Horst and Thoai (1997).

2.5. Weak Pareto Optimality

In addition to Pareto optimality, other related concepts are widely used. These are weak and proper Pareto optimality. The relationship between these concepts is that the properly Pareto optimal set is a subset of the Pareto optimal set which is a subset of the weakly Pareto optimal set.

A vector is weakly Pareto optimal if there does not exist any other vector for which all the components are better. More formally it means the following:

Definition 2.5.1. A decision vector $\mathbf{x}^* \in S$ is *weakly Pareto optimal* if there does not exist another decision vector $\mathbf{x} \in S$ such that $f_i(\mathbf{x}) < f_i(\mathbf{x}^*)$ for all $i = 1, \ldots, k$.

An objective vector $\mathbf{z}^* \in Z$ is weakly Pareto optimal if there does not exist another objective vector $\mathbf{z} \in Z$ such that $z_i < z_i^*$ for all $i = 1, \ldots, k$; or equivalently, if the decision vector corresponding to it is weakly Pareto optimal.

The bold line in Figure 2.5.1 represents the set of weakly Pareto optimal objective vectors. The fact that the Pareto optimal set is a subset of the weakly Pareto optimal set can also be seen in the figure. The Pareto optimal objective vectors are situated along the line between the dots.

Similarly to Pareto optimality, local weak Pareto optimality can be defined in addition to the global weak Pareto optimality of Definition 2.5.1. It must still be kept in mind that usually only locally weakly Pareto optimal solutions are computationally available. Nevertheless, for the sake of brevity, we shall usually refer only to weak Pareto optimality.

Let us state as a curiosity that if the feasible region is convex and the objective functions are quasiconvex with at least one strictly quasiconvex function, the set of locally Pareto optimal solutions is a subset of the set of weakly Pareto

It is important to realize that regardless of the existence of an underlying value function, a general assumption still is that that less is preferred to more by the decision maker, that is, lower objective function values are preferred to higher. This assumption is usually made even in methods not involving value functions in any way. Thus, assuming that less is preferred to more is a more general assumption than assuming a strongly decreasing value function.

2.7. Efficiency

It is possible to define optimality in a multiobjective context in more general ways than by Pareto or weak Pareto optimality. Let us have a pointed convex cone D defined in $\mathbf{R}^k$. This cone D is called an *ordering cone* and it is used to induce a partial ordering on Z. Let us have two objective vectors, $\mathbf{z}^1$ and $\mathbf{z}^2 \in Z$. An objective vector $\mathbf{z}^1$ *dominates* $\mathbf{z}^2$, denoted by $\mathbf{z}^1 \leq_D \mathbf{z}^2$, if $\mathbf{z}^2 - \mathbf{z}^1 \in D$ and $\mathbf{z}^1 \neq \mathbf{z}^2$, that is, $\mathbf{z}^2 - \mathbf{z}^1 \in D \setminus \{\mathbf{0}\}$. The same can also be written as $\mathbf{z}^2 \in \mathbf{z}^1 + D$ and $\mathbf{z}^1 \neq \mathbf{z}^2$, that is, $\mathbf{z}^2 \in \mathbf{z}^1 + D \setminus \{\mathbf{0}\}$ as illustrated in Figure 2.7.1.

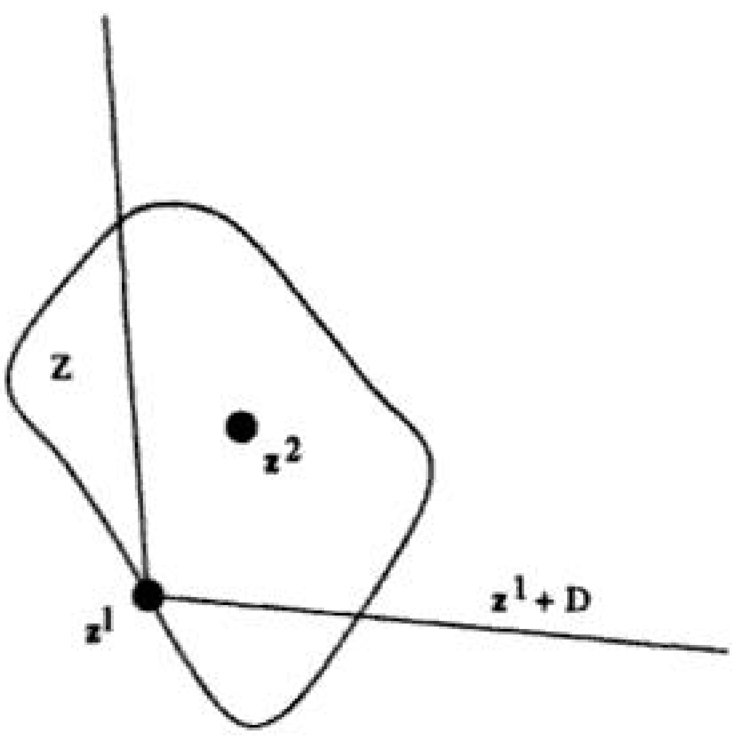

Figure 2.7.1. Domination induced by a cone D.

We can now present a definition of optimality based on domination, which is an alternative to the definitions previously given. When an ordering cone is used in defining optimality, then the term efficiency will be used in what follows.

Definition 2.7.1. Let D be a pointed convex cone. A decision vector $\mathbf{x}^* \in S$ is *efficient* (with respect to D) if there does not exist another decision vector $\mathbf{x} \in S$ such that $\mathbf{f}(\mathbf{x}) \leq_D \mathbf{f}(\mathbf{x}^*)$.

An objective vector $\mathbf{z}^* \in Z$ is efficient if there does not exist another objective vector $\mathbf{z} \in Z$ such that $\mathbf{z} \leq_D \mathbf{z}^*$.

Definition 2.8.4. Let the objective functions be continuously differentiable at a decision vector $\mathbf{x}^* \in S$. Then a partial trade-off rate at $\mathbf{x}^*$, involving f_i and f_j, is given by

$$\lambda_{ij} = \lambda_{ij}(\mathbf{x}^*) = \frac{\partial f_i(\mathbf{x}^*)}{\partial f_j}.$$

Differing from the idea of the definitions above, a so-called global trade-off is defined in Kaliszewski and Michalowski (1995, 1997). A global trade-off involves two objective functions and one decision vector which does not have to be Pareto optimal. It is the largest pairwise trade-off of two objective functions for one decision vector. Let us consider $\mathbf{x}^* \in S$ and modify the definitions for minimization problems. We define a subset of the feasible decision vectors in the form

$$S_j^>(\mathbf{x}^*) = \{\mathbf{x} \in S \mid f_j(\mathbf{x}) > f_j(\mathbf{x}^*),\ f_i(\mathbf{x}) \le f_i(\mathbf{x}^*),\ \text{ for } i = 1, \ldots, k,\ i \ne j\}.$$

Now we can introduce global trade-offs.

Definition 2.8.5. (From Kaliszewski and Michalowski (1995, 1997)) Let $\mathbf{x}^* \in S$ be a decision vector. We denote a *global trade-off* between the functions f_i and f_j by

$$\Lambda_{ij}^G = \Lambda_{ij}^G(\mathbf{x}^*) = \sup_{\mathbf{x} \in S_j^>(\mathbf{x}^*)} \frac{f_i(\mathbf{x}^*) - f_i(\mathbf{x})}{f_j(\mathbf{x}) - f_j(\mathbf{x}^*)}.$$

If $S_j^>(\mathbf{x}^*) = \emptyset$, then $\Lambda_{ij}^G(\mathbf{x}^*) = -\infty$ for every $i = 1, \ldots, k$, $i \ne j$.

A generalized definition of trade-offs in terms of tangent cones, meaning feasible directions, in the objective space is presented in Henig and Buchanan (1994, 1997). These generalized trade-off directions can be used for calculating trade-off rates at every Pareto optimal point of a convex multiobjective optimization problem.

Note that trade-offs are defined mathematically and the decision maker cannot affect them. If we take into consideration the opinions of the decision maker, we can define indifference curves and marginal rates of substitution.

2.8.2. Marginal Rate of Substitution

It is said that two feasible solutions are situated on the same *indifference curve* (or isopreference curve) if the decision maker finds them equally desirable, that is, neither of them is preferred to the other one. This means that indifference curves are contours of the underlying value function. There may also be a 'wider' *indifference band*. In this case we do not have any well-defined boundary between preferences, but a band where indifference occurs. This concept is studied in Passy and Levanon (1984).

Note that this definition differs from that of Pareto optimality so that a larger set $\mathbf{R}^k_\varepsilon$ is used instead of the set $\mathbf{R}^k_+$. The set of ε-properly Pareto optimal solutions is depicted in Figure 2.9.1 and denoted by a bold line. The solutions are obtained by intersecting the feasible objective region with a blunt cone. The end points of the Pareto optimal set, $\mathbf{z}^1$ and $\mathbf{z}^2$, have also been marked to ease the comparison.

An alternative formulation of Definition 2.9.2 is that a decision vector $\mathbf{x}^* \in S$ and the corresponding $\mathbf{z}^* \in Z$ are ε-properly Pareto optimal if $(\mathbf{z}^* - \mathbf{R}^k_\varepsilon) \cap Z = \mathbf{z}^*$. (The definition can be generalized into proper efficiency by using a convex cone D such that $\mathbf{R}^k_+ \subset \operatorname{int} D \cup \{\mathbf{0}\}$.)

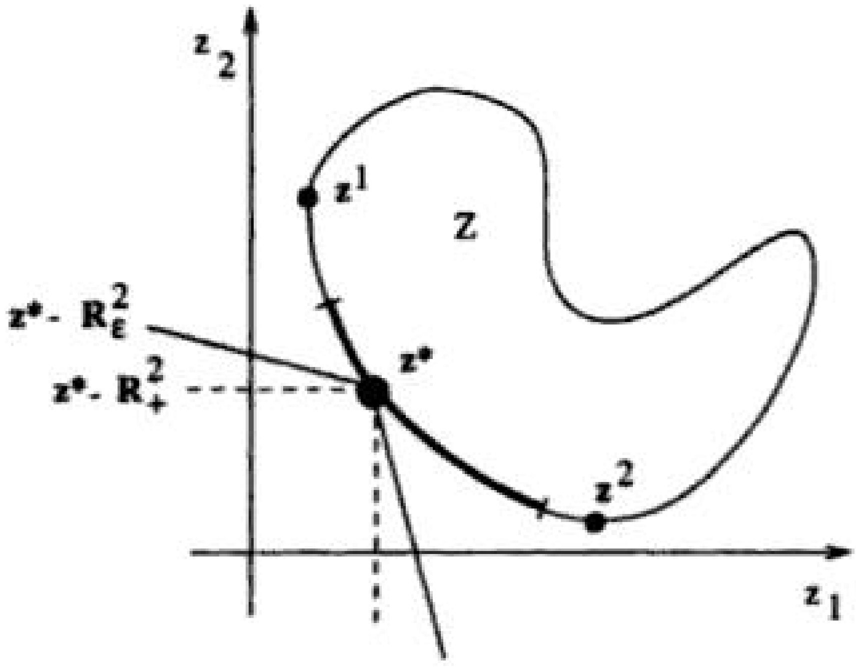

Figure 2.9.1. The set of ε-properly Pareto optimal solutions.

An interesting aspect of ε-properly Pareto optimal solutions is that the trade-offs are bounded by ε and $1/\varepsilon$ (see Wierzbicki (1986a, b)). We return to this concept in Section 3.5 of Part II.

Before we continue with the original definition of Kuhn and Tucker, we should mention briefly another way of decreasing the set of Pareto optimal solutions according to Liu (1996). There, $\mathbf{z}^* \in Z$ is called ϵ-Pareto optimal if $(\mathbf{z}^* - (\mathbf{R}^k_+ + \boldsymbol{\epsilon}) \setminus \{\mathbf{0}\}) \cap Z = \emptyset$, where $\boldsymbol{\epsilon} \in \mathbf{R}^k_+$.

Let us for a while assume that the feasible region is defined with the help of inequality constraints. In other words, $S = \{\mathbf{x} \in \mathbf{R}^n \mid \mathbf{g}(\mathbf{x}) = (g_1(\mathbf{x}), g_2(\mathbf{x}), \ldots, g_m(\mathbf{x}))^T \leq \mathbf{0}\}$. In addition, all the objective and the constraint functions are assumed to be continuously differentiable at every point $\mathbf{x} \in S$. Thus the next definition is not applicable to nondifferentiable multiobjective optimization problems.

Definition 2.9.3. (From Kuhn and Tucker (1951)) A decision vector $\mathbf{x}^* \in S$ is *properly Pareto optimal (in the sense of Kuhn and Tucker)* if it is Pareto optimal and if there does not exist any vector $\mathbf{d} \in \mathbf{R}^n$ such that

$$\nabla f_i(\mathbf{x}^*)^T \mathbf{d} \leq 0$$

(4) If in addition to the conditions in (3), the set $\{\hat{\mathbf{z}} \in \mathbf{R}^k \mid \hat{\mathbf{z}} \leq \mathbf{f}(\mathbf{x})$ for some $\mathbf{x} \in S\}$ is closed, then the Pareto optimal set is empty.

Proof. See Benson (1978) or Chankong and Haimes (1983b, pp. 151–152).

Problem (2.10.2) is a popular way of checking Pareto optimality and of generating Pareto optimal solutions. However, sometimes equality constraints cause computational difficulties. Therefore it is useful to note that the equalities in (2.10.2) can be replaced with inequalities $f_i(\mathbf{x}) + \varepsilon_i \leq f_i(\mathbf{x}^*)$ for all $i = 1, \ldots, k$ without affecting the generality of the results presented.

Two simple tests are suggested in Brosowski and da Silva (1994) for determining whether a given point is (locally) Pareto optimal or not. The tests are not based on any scalarizing functions but linear systems of equations. There are, however, several limitations. The objective functions are assumed to be continuously differentiable and their number has to be strictly larger than the number of variables. Further, no constraints can be included. Finally, the tests may also fail as demonstrated in Brosowski and da Silva (1994).

It is proved in Sawaragi et al. (1985, p. 59), that Pareto optimal solutions exist to multiobjective optimization problems where all the objective functions are lower semicontinuous (more general than continuity) and the feasible region is compact. Several ways of determining the Pareto optimality of a particular point in an MOLP problem are presented in Eiselt et al. (1987). They all apply to special situations. Further, the existence of Pareto optimal solutions when there is an infinite number of objective functions is considered in Alekseichik and Naumov (1981).

The existence of weakly Pareto optimal solutions in convex differentiable multiobjective optimization problems is treated in Deng (1998a). In addition, the compactness of the weakly Pareto optimal set is considered. The nonemptiness of the Pareto optimal and the weakly Pareto optimal sets in convex problems is also characterized in Deng (1998b).

As mentioned, auxiliary problems (2.10.1) and (2.10.2) can be used to produce Pareto optimal solutions, for example, from weakly Pareto optimal solutions. However, in some practical problems it is very expensive to carry out these additional optimizations. An alternative is suggested in Helbig (1991). If optimality is defined by an ordering cone, efficient solutions can be generated by perturbing this cone. In other words, using a method producing weakly efficient solutions with respect to the perturbed cone gives results that are efficient vis a vis the original problem.

The existence and the characterization of efficient solutions with respect to ordering cones are studied in Henig (1982a), and the existence of efficient solutions in linear spaces is treated in Borwein (1983). In addition, the existence of weakly and properly efficient (in the sense of Borwein) and efficient solutions in the presence of ordering cones is studied in Jahn (1986b). The existence of

Before we can continue, we write down the so-called Motzkin's theorem of the alternative. It will be needed in the proof of the following necessary condition.

Theorem 3.1.4. (*Motzkin's theorem*) Let A and C be given matrices. Then either the system of inequalities

$$A\mathbf{x} < \mathbf{0}, \quad C\mathbf{x} \leq \mathbf{0}$$

has a solution $\mathbf{x}$, or the system

$$A^T\boldsymbol{\lambda} + C^T\boldsymbol{\mu} = 0, \quad \boldsymbol{\lambda} \geq \mathbf{0}, \quad \boldsymbol{\lambda} \neq \mathbf{0}, \quad \boldsymbol{\mu} \geq \mathbf{0}$$

has a solution $(\boldsymbol{\lambda}, \boldsymbol{\mu})$, but never both.

Proof. See, for example, Mangasarian (1969, pp. 28–29).

Now we can formulate the Karush-Kuhn-Tucker necessary condition for Pareto optimality.

Theorem 3.1.5. (*Karush-Kuhn-Tucker necessary condition for Pareto optimality*) Let the assumptions of Theorem 3.1.1 be satisfied by the Kuhn-Tucker constraint qualification. Theorem 3.1.1 is then valid with the addition that $\boldsymbol{\lambda} \neq \mathbf{0}$.

Proof. Let $\mathbf{x}^* \in S$ be Pareto optimal. The idea of this proof is to apply Theorem 3.1.4. For this reason we prove that there does not exist any $\mathbf{d} \in \mathbf{R}^n$ such that

$$\begin{aligned} &\nabla f_i(\mathbf{x}^*)^T\mathbf{d} < 0 \text{ for all } i = 1, \ldots, k, \text{ and} \\ &\nabla g_j(\mathbf{x}^*)^T\mathbf{d} \leq 0 \text{ for all } j \in J(\mathbf{x}^*). \end{aligned} \tag{3.1.2}$$

Let us on the contrary assume that there exists some $\mathbf{d}^* \in \mathbf{R}^n$ satisfying (3.1.2). Then from the Kuhn-Tucker constraint qualification we know that there exists a function $\mathbf{a}\colon [0, 1] \to \mathbf{R}^n$ which is continuously differentiable at 0 and some real scalar $\alpha > 0$ such that $\mathbf{a}(0) = \mathbf{x}^*$, $\mathbf{g}(\mathbf{a}(t)) \leq \mathbf{0}$ for all $0 \leq t \leq 1$ and $\mathbf{a}'(0) = \alpha\mathbf{d}^*$.

Because the functions f_i are continuously differentiable, we can approximate $f_i(\mathbf{a}(t))$ linearly as

$$\begin{aligned} f_i(\mathbf{a}(t)) &= f_i(\mathbf{x}^*) + \nabla f_i(\mathbf{x}^*)^T(\mathbf{a}(t) - \mathbf{x}^*) + \|\mathbf{a}(t) - \mathbf{x}^*\|\varphi(\mathbf{a}(t), \mathbf{x}^*) \\ &= f_i(\mathbf{x}^*) + \nabla f_i(\mathbf{x}^*)^T(\mathbf{a}(t) - \mathbf{a}(0)) + \|\mathbf{a}(t) - \mathbf{a}(0)\|\varphi(\mathbf{a}(t), \mathbf{a}(0)) \\ &= f_i(\mathbf{x}^*) + t\nabla f_i(\mathbf{x}^*)^T\left(\frac{\mathbf{a}(0 + t) - \mathbf{a}(0)}{t}\right) + \|\mathbf{a}(t) - \mathbf{a}(0)\|\varphi(\mathbf{a}(t), \mathbf{a}(0)), \end{aligned}$$

where $\varphi(\mathbf{a}(t), \mathbf{a}(0)) \to 0$ as $\|\mathbf{a}(t) - \mathbf{a}(0)\| \to 0$. As $t \to 0$ tends $\|\mathbf{a}(t) - \mathbf{a}(0)\|$ to zero and $(\mathbf{a}(0 + t) - \mathbf{a}(0))/t \to \mathbf{a}'(0) = \alpha\mathbf{d}^*$.

is the kind of a second-order constraint qualification that can guarantee an even stronger result where not all the λ-coefficients can vanish.

The following theorem gives two second-order sufficient optimality conditions. The difference lies in the sets of search directions.

Theorem 3.1.12. (*Second-order sufficient condition for Pareto optimality*) Let the objective and the constraint functions of problem (3.1.1) be twice continuously differentiable at a decision vector $\mathbf{x}^* \in S$. A sufficient condition for $\mathbf{x}^*$ to be Pareto optimal is that there exist vectors $\mathbf{0} \leq \boldsymbol{\lambda} \in \mathbf{R}^k$ and $\mathbf{0} \leq \boldsymbol{\mu} \in \mathbf{R}^m$ for which $(\boldsymbol{\lambda}, \boldsymbol{\mu}) \neq (\mathbf{0}, \mathbf{0})$ such that

$$(1) \quad \sum_{i=1}^{k} \lambda_i \nabla f_i(\mathbf{x}^*) + \sum_{j=1}^{m} \mu_j \nabla g_j(\mathbf{x}^*) = 0$$

$$(2) \quad \mu_j g_j(\mathbf{x}^*) = 0 \quad \text{for all} \quad j = 1, \ldots, m$$

$$(3) \quad \mathbf{d}^T \Big(\sum_{i=1}^{k} \lambda_i \nabla^2 f_i(\mathbf{x}^*) + \sum_{j=1}^{m} \mu_j \nabla^2 g_j(\mathbf{x}^*) \Big) \mathbf{d} > 0$$

for either all $\mathbf{d} \in \{\mathbf{0} \neq \mathbf{d} \in \mathbf{R}^n \mid \nabla f_i(\mathbf{x}^*)^T \mathbf{d} \leq 0 \text{ for all } i = 1, \ldots, k,\ \nabla g_j(\mathbf{x}^*)^T \mathbf{d} \leq 0 \text{ for all } j \in J(\mathbf{x}^*)\}$ or all $\mathbf{d} \in \{\mathbf{0} \neq \mathbf{d} \in \mathbf{R}^n \mid \nabla g_j(\mathbf{x}^*)^T \mathbf{d} = 0 \text{ for all } j \in J^+(\mathbf{x}^*),\ \nabla g_j(\mathbf{x}^*)^T \mathbf{d} \leq 0 \text{ for all } j \in J(\mathbf{x}^*) \setminus J^+(\mathbf{x}^*)\}$, where $J^+(\mathbf{x}^*) = \{j \in J(\mathbf{x}^*) \mid \mu_j > 0\}$.

Proof. See Wang (1991).

Second-order sufficient conditions for Pareto optimality are also treated in Simon (1986), and more necessary and sufficient conditions for Pareto and weakly Pareto optimal solutions are presented in Wang (1991).

3.1.3. Conditions for Proper Pareto Optimality

For completeness we also present the original necessary optimality condition formulated for proper Pareto optimality in the sense of Kuhn and Tucker (see Definition 2.9.3) as stated by Kuhn and Tucker (1951). To begin with, we write down Tucker's theorem of the alternative, which will be utilized in the proof.

Theorem 3.1.13. (*Tucker's theorem*) Let A and C be given matrices. Then either the system of inequalities

$$A\mathbf{x} \leq \mathbf{0}, \quad A\mathbf{x} \neq \mathbf{0}, \quad C\mathbf{x} \leq \mathbf{0}$$

has a solution $\mathbf{x}$, or the system

$$A^T \boldsymbol{\lambda} + C^T \boldsymbol{\mu} = 0, \quad \boldsymbol{\lambda} > \mathbf{0}, \quad \boldsymbol{\mu} \geq \mathbf{0}$$

has a solution $(\boldsymbol{\lambda}, \boldsymbol{\mu})$, but never both.

3.2.1. First-Order Conditions

The first result to be presented is a necessary condition of the Fritz John type for Pareto optimality.

Theorem 3.2.5. (*Fritz John necessary condition for Pareto optimality*) Let the objective and the constraint functions of problem (3.2.1) be locally Lipschitzian at a point $\mathbf{x}^* \in S$. A necessary condition for the point $\mathbf{x}^*$ to be Pareto optimal is that there exist multipliers $\mathbf{0} \leq \boldsymbol{\lambda} \in \mathbf{R}^k$ and $\mathbf{0} \leq \boldsymbol{\mu} \in \mathbf{R}^m$ for which $(\boldsymbol{\lambda}, \boldsymbol{\mu}) \neq (\mathbf{0}, \mathbf{0})$ such that

$$(1) \quad 0 \in \sum_{i=1}^{k} \lambda_i \partial f_i(\mathbf{x}^*) + \sum_{j=1}^{m} \mu_j \partial g_j(\mathbf{x}^*)$$

$$(2) \quad \mu_j g_j(\mathbf{x}^*) = 0 \text{ for all } j = 1, \ldots, m.$$

Proof. Because it is assumed that $(\boldsymbol{\lambda}, \boldsymbol{\mu}) \neq (\mathbf{0}, \mathbf{0})$, we can normalize the multipliers to sum up to one. We shall here prove a stronger condition, where $\sum_{i=1}^{k} \lambda_i + \sum_{j=1}^{m} \mu_j = 1$.

Let $\mathbf{x}^* \in S$ be Pareto optimal. At first we define an additional function $F: \mathbf{R}^n \to \mathbf{R}$ by

$$F(\mathbf{x}) = \max\,[f_i(\mathbf{x}) - f_i(\mathbf{x}^*), g_j(\mathbf{x}) \mid i = 1, \ldots, k,\ j = 1, \ldots, m]$$

and show that for all $\mathbf{x} \in \mathbf{R}^n$ is

$$(3.2.2) \qquad F(\mathbf{x}) \geq 0.$$

Let us on the contrary assume that for some $\mathbf{x}^\circ \in \mathbf{R}^n$ is $F(\mathbf{x}^\circ) < 0$. Then $g_j(\mathbf{x}^\circ) < 0$ for all $j = 1, \ldots, m$ and the point $\mathbf{x}^\circ$ is thus feasible in problem (3.2.1). In addition, $f_i(\mathbf{x}^\circ) < f_i(\mathbf{x}^*)$ for all $i = 1, \ldots, k$, which contradicts the Pareto optimality of $\mathbf{x}^*$. Thus (3.2.2) must be true.

Noting that the point $\mathbf{x}^*$ is feasible in problem (3.2.1), we obtain $\mathbf{g}(\mathbf{x}^*) \leq \mathbf{0}$. This implies $F(\mathbf{x}^*) = 0$. Combining this fact with property (3.2.2), we know that F attains its (global) minimum at $\mathbf{x}^*$. As all the functions f_i and g_j are locally Lipschitzian at $\mathbf{x}^*$, likewise F (according to Theorem 3.2.2). We deduce from Theorem 3.2.3 that $0 \in \partial F(\mathbf{x}^*)$.

Note that

$$(3.2.3) \qquad \partial(f_i(\mathbf{x}) - f_i(\mathbf{x}^*)) = \partial f_i(\mathbf{x}),$$

applying Theorem 3.2.1.

We designate the set of indices j for which $F(\mathbf{x}^*) = g_j(\mathbf{x}^*)$ by $J(\mathbf{x}^*) \subset \{1, \ldots, m\}$. Now we can employ Theorem 3.2.2 and (3.2.3) and obtain

$$0 \in \operatorname{conv}\{\partial f_i(\mathbf{x}^*), \partial g_j(\mathbf{x}^*) \mid i = 1, \ldots, k,\ j \in J(\mathbf{x}^*)\}.$$

Theorem 3.2.12. (*Karush-Kuhn-Tucker sufficient condition for weak Pareto optimality*) The condition stated in Theorem 3.2.11 is sufficient for a decision vector $\mathbf{x}^* \in S$ to be weakly Pareto optimal for $\mathbf{0} \leq \boldsymbol{\lambda} \in \mathbf{R}^k$ with $\boldsymbol{\lambda} \neq \mathbf{0}$.

Proof. The proof is a trivial modification of the proof of Theorem 3.2.11.

Finally, we introduce one more constraint qualification. It can only be applied to convex problems and it will also be needed in Part II (Section 2.2) in connection with the multiobjective proximal bundle method. It is called the Slater constraint qualification. It is independent of the differentiability of the functions involved.

Definition 3.2.13. Let problem (3.2.1) be convex. Problem (3.2.1) satisfies the *Slater constraint qualification* if there exists some $\mathbf{x}$ with $g_j(\mathbf{x}) < 0$ for all $j = 1, \ldots, m$.

Theorem 3.2.9 and Corollary 3.2.10 can now be reformulated for convex problems assuming the Slater constraint qualification. Remember that convexity implies that functions are locally Lipschitzian at any point in the feasible region.

Theorem 3.2.14. (*Karush-Kuhn-Tucker necessary condition for (weak) Pareto optimality*) Let problem (3.2.1) be convex, satisfying the Slater constraint qualification. A necessary condition for a point $\mathbf{x}^* \in S$ to be (weakly) Pareto optimal is that there exist multipliers $\mathbf{0} \leq \boldsymbol{\lambda} \in \mathbf{R}^k$ with $\boldsymbol{\lambda} \neq \mathbf{0}$ and $\mathbf{0} \leq \boldsymbol{\mu} \in \mathbf{R}^m$ such that

$$(1) \quad 0 \in \sum_{i=1}^{k} \lambda_i \partial f_i(\mathbf{x}^*) + \sum_{j=1}^{m} \mu_j \partial g_j(\mathbf{x}^*)$$

$$(2) \quad \mu_j g_j(\mathbf{x}^*) = 0 \quad \text{for all} \quad j = 1, \ldots, m.$$

Proof. The proof is a trivial modification of the proof of Theorem 3.2.9 when we note the following. In case the set $J(\mathbf{x}^*)$ is nonempty we denote $g(\mathbf{x}) = \max\,[g_j(\mathbf{x}) \mid j = 1, \ldots, m]$. Now $g(\mathbf{x}^*) = g_j(\mathbf{x}^*)$ for $j \in J(\mathbf{x}^*)$. By the Slater constraint qualification there exists some $\mathbf{x}^\circ$ such that $g_j(\mathbf{x}^\circ) < 0$ for all j. Thus, $\mathbf{x}^*$ cannot be the global minimum of the convex function g. According to Theorem 3.2.3 we derive

$$0 \notin \operatorname{conv}\{\partial g_j(\mathbf{x}^*) \mid j \in J(\mathbf{x}^*)\}.$$

The proof of Theorem 3.2.9 can now be applied. □

Necessary optimality conditions for Pareto optimality in those nondifferentiable problems where the objective functions are fractions of convex and

In Chankong and Haimes (1982), the Karush-Kuhn-Tucker optimality conditions for Pareto optimality are modified for use in connection with certain solution methods (the ε-constraint method and the jth Lagrangian problem; see Section 3.2 of Part II). Chankong and Haimes also propose optimality conditions for proper Pareto optimality (in the sense of Geoffrion) with the ε-constraint method. Further, in Benson and Morin (1977), necessary and sufficient conditions are given for a Pareto optimal solution to be properly Pareto optimal. This is done with the help of the jth Lagrangian problem. Necessary and sufficient conditions for Pareto optimality with convex and differentiable functions partly based on the ε-constraint problem are proved in Zlobec (1984).

Necessary and sufficient conditions for Pareto optimality and proper Pareto optimality are proved with the help of duality theory and auxiliary problem (2.10.1) (presented in Section 2.10) in Wendell and Lee (1977). However, it is stated that nonlinear problems do not generally satisfy the conditions developed. In such cases Pareto optimal solutions have to be tested for proper Pareto optimality on a point-by-point basis.

In Gulati and Islam (1988), linear fractional objective functions and generalized convex constraints are handled. Necessary conditions of the Karush-Kuhn-Tucker type are presented for Pareto optimal solutions, and the conditions under which Pareto optimal solutions are properly Pareto optimal are stated. Necessary and sufficient conditions for Pareto optimality in problems with nonlinear fractional objective functions and nonlinear constraints are proved in Lee (1992). In addition, necessary optimality conditions for fractional multiobjective optimization problems with square root terms are given in Egudo (1991). In Benson (1979b), a necessary and sufficient condition is given for a point to be Pareto optimal when there are two concave objective functions (problem of maximization) and a convex feasible set.

The following references deal with conditions for efficiency, where the objective space is ordered by an ordering cone.

In Zubiri (1988), necessary and sufficient conditions are proved for weak efficiency in Banach spaces with the help of a weighted L_∞-metric (see Section 3.4 of Part II). Several necessary and sufficient conditions for efficient, weakly efficient and properly efficient solutions (in the sense of Borwein) in real topological linear spaces are collected in Jahn (1985). Necessary and sufficient optimality conditions of the Karush-Kuhn-Tucker type are derived in Hazen (1988), for cases where preferences are and are not representable by cones.

Let us finally briefly mention some further references handling nondifferentiable cases. Necessary and sufficient conditions for Pareto optimality and proper Pareto optimality are derived in Bhatia and Aggarwal (1992), by the weighting method (see Section 3.1 of Part II) and Dini derivatives. The functions in the problem are assumed to be nondifferentiable such that the objective functions are pseudoconcave and the constraint functions are quasiconvex. Some duality results are provided as well.

1. INTRODUCTION

Generating Pareto optimal solutions plays an important role in multiobjective optimization, and mathematically the problem is considered to be solved when the Pareto optimal set is found. The term *vector optimization* is sometimes used to denote the problem of identifying the Pareto optimal set. However, this is not always enough. We want to obtain only one solution. This means that we must find a way to put the Pareto optimal solutions in a complete order. This is why we need a decision maker and her or his preference structure. Here in Part II, we present several methods for solving multiobjective optimization problems. Usually, this means finding the Pareto optimal solution that best satisfies the decision maker.

We are not here going to interfere with the formulation of a real-life phenomenon as a mathematically well-defined problem. We merely stress that a proper formulation is important. Let us emphasize that in real-life problems inaccuracy in some form is often present. Remember that we exclude the handling of stochastic or fuzzy problems in this context. Even when the problems are modelled in a deterministic form, restricting the treatment to Pareto optimal solutions only may be misleading. For example, forgetting or misspelling an objective function may affect the Pareto optimal set. If it is impossible to model the practical problem in an explicit and precise mathematical form, we cannot automatically leave non-Pareto optimal solutions out of consideration. For example, imprecision of the data, the measurement or the objective functions means that the Pareto optimal set available is only an approximation of the real one. Here we have a gap between theory and practice.

Several crucial issues to bear in mind in the formulation of problems are treated in Haimes (1985) and Nijkamp et al. (1988). Among these are risk assessment, sufficient representativeness of the objective functions and precision of information. In many complicated, practical cases it may be impossible to give a correct formulation to the problem before it is solved. This means that the modelling and the solution phases should not be undertaken separately, which is generally the case nowadays. In other words, the modelling phase may require interaction with the solution phase. A parallel idea of approaching the modelling phase by including the decision maker in the modelling is suggested in Brans (1996). The goal is to give more freedom to the decision maker and not to limit her or his way of thinking to a prespecified model and its concepts.

(1973). A wide collection of methods available (up to the year 1983) is assembled also in Despontin et al. (1983). Almost 100 methods for both multiobjective and multiattribute cases are included.

As far as different nationalities are concerned, overviews of multiobjective optimization methods in the former Soviet Union are presented in Lieberman (1991a, b) and of theory and applications in China in Hu (1990). Nine multiobjective optimization methods developed in Germany are briefly introduced in Ester and Holzmüller (1986).

A great number of interactive multiobjective optimization methods is collected in Shin and Ravindran (1991) and Vanderpooten and Vincke (1989). Interactive methods are also presented in Narula and Weistroffer (1989a) and White (1983b). Information about applications of the methods is also reported. Some literature on interactive multiobjective optimization between the years 1965 and 1988 is gathered in Aksoy (1990). A set of scalarizing functions is outlined in Wierzbicki (1986b) with special attention to whether weakly, properly or Pareto optimal solutions are produced.

As to different problem types, an overview of methods for MOLP problems can be found in Zionts (1980, 1989). Methods for hierarchical multiobjective optimization problems are reviewed in Haimes and Li (1988). Such methods are needed in large-scale problems. A wide survey on the literature of hierarchical multiobjective analysis is also provided.

Methods with applications to large-scale systems and industry are presented in the monographs Haimes et al. (1990) and Tabucanon (1988), respectively. Several groups of methods applicable to computer-aided design systems are presented briefly in Eiduks (1983). Methods for applications in structural optimization are reported in Eschenauer (1987), Jendo (1986), Koski and Silvennoinen (1987) and Osyczka and Koski (1989). The collections of papers edited by Eschenauer et al. (1990a) and Stadler (1988a) contain mainly applications in engineering.

In the following, we present several methods (in four classes) for multiobjective optimization. Some of them will be described in more detail and some only briefly mentioned. It must be kept in mind that the existing methodology is very wide. We do not intend to cover every existing method but to introduce several philosophies and ways of approaching multiobjective optimization problem solving. Where possible we try to link references to some of the applications and extensions available in the literature with the methods presented here. The description of each method ends with concluding remarks by the author taking up important aspects of the method. Unless stated otherwise, we assume that we solve problem (2.1.1) defined in Part I.

In connection with methods, a mention is made only of such implementations as have been made available to the author for testing purposes. By a *user* we mean either a decision maker or an analyst who uses the solution program. If the user is a decision maker, it is usually assumed that the problem has been formulated earlier (and perhaps loaded in the system) so that the decision maker can concentrate on the actual solution process.

Theorem 2.1.1. The solution of L_p-problem (2.1.1) (where $1 \leq p < \infty$) is Pareto optimal.

Proof. Let $\mathbf{x}^* \in S$ be a solution of problem (2.1.1) with $1 \leq p < \infty$. Let us suppose that $\mathbf{x}^*$ is not Pareto optimal. Then there exists a point $\mathbf{x} \in S$ such that $f_i(\mathbf{x}) \leq f_i(\mathbf{x}^*)$ for all $i = 1, \ldots, k$ and $f_j(\mathbf{x}) < f_j(\mathbf{x}^*)$ for at least one j. Now $(f_i(\mathbf{x}) - z_i^\star)^p \leq (f_i(\mathbf{x}^*) - z_i^\star)^p$ for all i and $(f_j(\mathbf{x}) - z_j^\star)^p < (f_j(\mathbf{x}^*) - z_j^\star)^p$. From this we obtain

$$\sum_{i=1}^{k} \left(f_i(\mathbf{x}) - z_i^\star\right)^p < \sum_{i=1}^{k} \left(f_i(\mathbf{x}^*) - z_i^\star\right)^p.$$

When both sides of the inequality are raised into the power $1/p$ we have a contradiction to the assumption that $\mathbf{x}^*$ is a solution of problem (2.1.1). This completes the proof. □

Yu has pointed out in Yu (1973) that if Z is a convex set, then for $1 < p < \infty$ the solution of problem (2.1.1) is unique.

Theorem 2.1.2. The solution of Tchebycheff problem (2.1.2) is weakly Pareto optimal.

Proof. Let $\mathbf{x}^* \in S$ be a solution of problem (2.1.2). Let us suppose that $\mathbf{x}^*$ is not weakly Pareto optimal. In this case, there exists a point $\mathbf{x} \in S$ such that $f_i(\mathbf{x}) < f_i(\mathbf{x}^*)$ for all $i = 1, \ldots, k$. It means that, $f_i(\mathbf{x}) - z_i^\star < f_i(\mathbf{x}^*) - z_i^\star$ for all i. Thus, $\mathbf{x}^*$ cannot be a solution of problem (2.1.2). Here we have a contradiction which completes the proof. □

Theorem 2.1.3. Tchebycheff problem (2.1.2) has at least one Pareto optimal solution.

Proof. Let us suppose that none of the optimal solutions of problem (2.1.2) is Pareto optimal. Let $\mathbf{x}^* \in S$ be an optimal solution of problem (2.1.2). Since we assume that it is not Pareto optimal, there must exist a solution $\mathbf{x} \in S$ which is not optimal for problem (2.1.2) but for which $f_i(\mathbf{x}) \leq f_i(\mathbf{x}^*)$ for all $i = 1, \ldots, k$ and $f_j(\mathbf{x}) < f_j(\mathbf{x}^*)$ for at least one j.

We have now $f_i(\mathbf{x}) - z_i^\star \leq f_i(\mathbf{x}^*) - z_i^\star$ for all i with the strict inequality holding for at least one index j, and further $\max_i[f_i(\mathbf{x}) - z_i^\star] \leq \max_i[f_i(\mathbf{x}^*) - z_i^\star]$. Because $\mathbf{x}^*$ is an optimal solution of problem (2.1.2), $\mathbf{x}$ has to be an optimal solution, as well. This contradiction completes the proof. □

Corollary 2.1.4. If Tchebycheff problem (2.1.2) has a unique solution, it is Pareto optimal.

A linear numerical application example of the method is given in Hwang and Masud (1979, pp. 23–29). Sufficient conditions for the solution of an L_p-problem

$$\bar{f}_{i,j}(\mathbf{x}) = f_i(\mathbf{y}^j) + (\boldsymbol{\xi}^j_{f_i})^T(\mathbf{x} - \mathbf{y}^j) \quad \text{for all } i = 1, \ldots, k,\ j \in J^h \quad \text{and}$$
$$\bar{g}_{l,j}(\mathbf{x}) = g_l(\mathbf{y}^j) + (\boldsymbol{\xi}^j_{g_l})^T(\mathbf{x} - \mathbf{y}^j) \quad \text{for all } l = 1, \ldots, m,\ j \in J^h.$$

We can now define a convex piecewise linear approximation to the improvement function by

$$\hat{H}^h(\mathbf{x}) = \max\,[\bar{f}_{i,j}(\mathbf{x}) - f_i(\mathbf{x}^h),\ \bar{g}_{l,j}(\mathbf{x}) \mid i = 1, \ldots, k,\ l = 1, \ldots, m,\ j \in J^h]$$

and we get an approximation to (2.2.3) by

$$(2.2.4) \qquad \begin{array}{ll} \text{minimize} & \hat{H}^h(\mathbf{x}^h + \mathbf{d}) + \frac{1}{2}u^h\|\mathbf{d}\|^2 \\ \text{subject to} & \mathbf{d} \in \mathbf{R}^n, \end{array}$$

where $u^h > 0$ is an inner parameter to be updated automatically. The penalty term $\frac{1}{2}u^h\|\mathbf{d}\|^2$ is added to guarantee that there exists a solution to problem (2.2.4) and to keep the approximation local enough.

Notice that (2.2.4) is still a nondifferentiable problem, but due to its min-max-nature it is equivalent to the following differentiable quadratic problem with $\mathbf{d} \in \mathbf{R}^n$ and $v \in \mathbf{R}$ as variables:

$$(2.2.5) \qquad \begin{array}{ll} \text{minimize} & v + \frac{1}{2}u^h\|\mathbf{d}\|^2 \\ \text{subject to} & v \geq -\alpha^h_{f_i,j} + (\boldsymbol{\xi}^j_{f_i})^T\mathbf{d},\ i = 1, \ldots, k,\ j \in J^h \\ & v \geq -\alpha^h_{g_l,j} + (\boldsymbol{\xi}^j_{g_l})^T\mathbf{d},\ l = 1, \ldots, m,\ j \in J^h, \end{array}$$

where

$$\alpha^h_{f_i,j} = f_i(\mathbf{x}^h) - \bar{f}_{i,j}(\mathbf{x}^h),\ i = 1, \ldots, k,\ j \in J^h \quad \text{and}$$
$$\alpha^h_{g_l,j} = -\bar{g}_{l,j}(\mathbf{x}^h),\ l = 1, \ldots, m,\ j \in J^h$$

are so-called linearization errors.

In the nonconvex case, we replace the linearization errors by so-called subgradient locality measures:

$$\beta^h_{f_i,j} = \max\ [|\alpha^h_{f_i,j}|, \gamma_{f_i}\|\mathbf{x}^h - \mathbf{y}^j\|^2]$$
$$\beta^h_{g_l,j} = \max\ [|\alpha^h_{g_l,j}|, \gamma_{g_l}\|\mathbf{x}^h - \mathbf{y}^j\|^2],$$

where $\gamma_{f_i} \geq 0$ for $i = 1, \ldots, k$ and $\gamma_{g_l} \geq 0$ for $l = 1, \ldots, m$ are so-called distance measure parameters ($\gamma_{f_i} = 0$ if f_i is convex and $\gamma_{g_l} = 0$ if g_l is convex).

Let $(\mathbf{d}^h, v^h)$ be a solution of problem (2.2.5). Then the two-point line search strategy is carried out. It detects discontinuities in the gradients of the objective functions. Roughly speaking, we try to find a step-size $0 < t^h \leq 1$ such that $H(\mathbf{x}^h + t^h\mathbf{d}^h, \mathbf{x}^h)$ is minimal when $\mathbf{x}^h + t^h\mathbf{d}^h \in S$.

A line search algorithm (in Mäkelä and Neittaanmäki (1992, pp. 126–130)) is used to produce the step-sizes. The iteration is terminated when a predetermined accuracy is reached. The subgradient aggregation strategy due to Kiwiel (1985c) is used to bound the storage requirements (i.e., the size of the

3.1. Weighting Method

In the weighting method, presented, for example, in Gass and Saaty (1955) and Zadeh (1963), the idea is to associate each objective function with a weighting coefficient and minimize the weighted sum of the objectives. In this way, the multiple objective functions are transformed into a single objective function. We suppose that the weighting coefficients w_i are real numbers such that $w_i \geq 0$ for all $i = 1, \ldots, k$. It is also usually supposed that the weights are normalized, that is, $\sum_{i=1}^{k} w_i = 1$. To be more exact, the multiobjective optimization problem is modified into the following problem, to be called a *weighting problem*:

$$\begin{aligned} &\text{minimize} \quad \sum_{i=1}^{k} w_i f_i(\mathbf{x}) \\ &\text{subject to} \quad \mathbf{x} \in S, \end{aligned} \tag{3.1.1}$$

where $w_i \geq 0$ for all $i = 1, \ldots, k$ and $\sum_{i=1}^{k} w_i = 1$.

3.1.1. Theoretical Results

In the following, several theoretical results are presented concerning the weighting method.

Theorem 3.1.1. The solution of weighting problem (3.1.1) is weakly Pareto optimal.

Proof. Let $\mathbf{x}^* \in S$ be a solution of the weighting problem. Let us suppose that it is not weakly Pareto optimal. In this case, there exists a solution $\mathbf{x} \in S$ such that $f_i(\mathbf{x}) < f_i(\mathbf{x}^*)$ for all $i = 1, \ldots, k$. According to the assumptions set to the weighting coefficients, $w_j > 0$ for at least one j. Thus we have $\sum_{i=1}^{k} w_i f_i(\mathbf{x}) < \sum_{i=1}^{k} w_i f_i(\mathbf{x}^*)$. This is a contradiction to the assumption that $\mathbf{x}^*$ is a solution of the weighting problem. Thus $\mathbf{x}^*$ is weakly Pareto optimal. □

Theorem 3.1.2. The solution of weighting problem (3.1.1) is Pareto optimal if the weighting coefficients are positive, that is $w_i > 0$ for all $i = 1, \ldots, k$.

Proof. Let $\mathbf{x}^* \in S$ be a solution of the weighting problem with positive weighting coefficients. Let us suppose that it is not Pareto optimal. This means that there exists a solution $\mathbf{x} \in S$ such that $f_i(\mathbf{x}) \leq f_i(\mathbf{x}^*)$ for all $i = 1, \ldots, k$ and $f_j(\mathbf{x}) < f_j(\mathbf{x}^*)$ for at least one j. Since $w_i > 0$ for all $i = 1, \ldots, k$, we have $\sum_{i=1}^{k} w_i f_i(\mathbf{x}) < \sum_{i=1}^{k} w_i f_i(\mathbf{x}^*)$. This contradicts the assumption that $\mathbf{x}^*$ is a solution of the weighting problem and, thus, $\mathbf{x}^*$ must be Pareto optimal. □

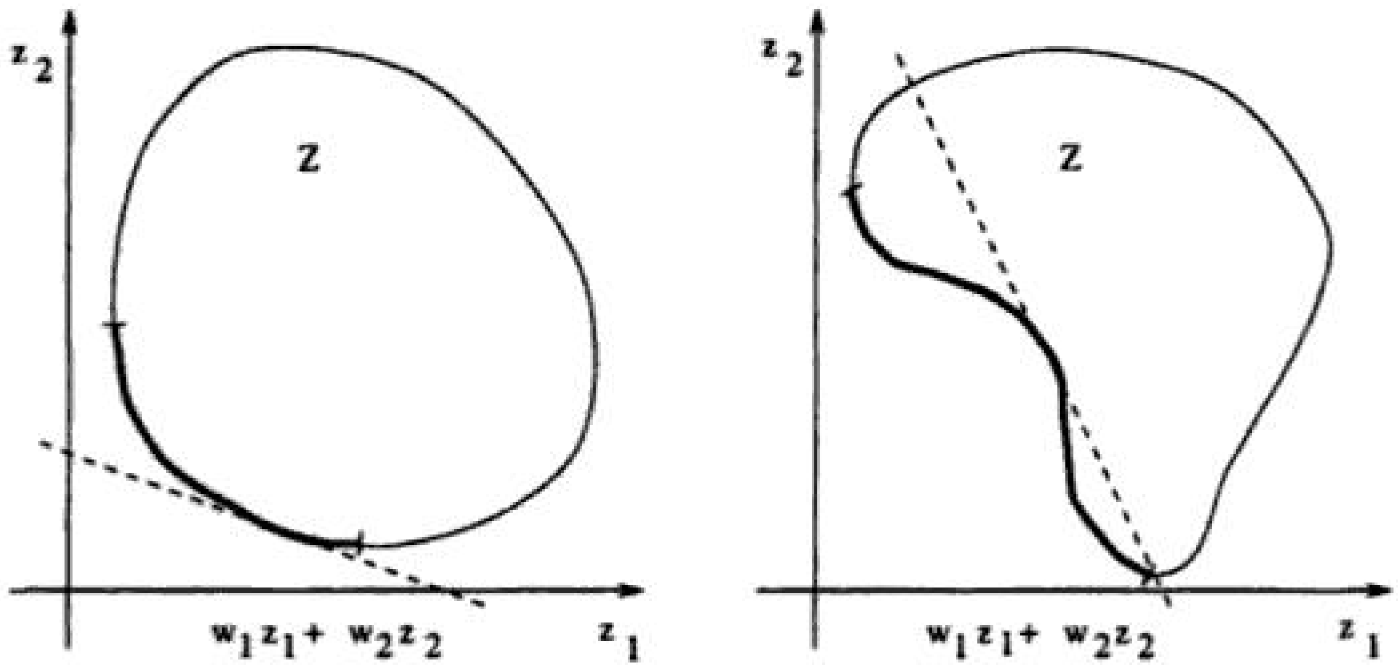

Figure 3.1.1. Weighting method with convex and nonconvex problems.

usually find only extreme point solutions. Thus, if some facet of the feasible region is Pareto optimal, then the infinity of Pareto optimal non-extreme points must be described in terms of linear combinations of the Pareto optimal extreme solutions. On the other hand, note that if two adjacent Pareto optimal extreme points for an MOLP problem are found, the edge connecting them is not necessarily Pareto optimal.

The conditions under which the whole Pareto optimal set can be generated by the weighting method with positive weighting coefficients are presented in Censor (1977). The solutions that it is possible to reach by the weighting method with positive weighting coefficients are characterized in Belkeziz and Pirlot (1991). Some generalized results are also given. More relations between nonnegative and positive weighting coefficients, convexity of S and Z, and Pareto optimality are studied in Lin (1976b).

If the weighting coefficients in the weighting method are all positive, we can say more about the solutions than that they are Pareto optimal. The following results concerning proper Pareto optimality were originally presented in Geoffrion (1968).

Theorem 3.1.6. The solution of weighting problem (3.1.1) is properly Pareto optimal if all the weighting coefficients are positive (sufficient condition).

Proof. Let $\mathbf{x}^* \in S$ be a solution of the weighting problem with positive weighting coefficients. In Theorem 3.1.2 we showed that the solution is Pareto optimal. We shall now show that $\mathbf{x}^*$ is properly Pareto optimal with $M = (k-1)\max_{i,j}(w_j/w_i)$.

Let us on the contrary suppose that $\mathbf{x}^*$ is not properly Pareto optimal. Then for some i (which we fix) and for $\mathbf{x} \in S$ such that $f_i(\mathbf{x}^*) > f_i(\mathbf{x})$ we have

$$f_i(\mathbf{x}^*) - f_i(\mathbf{x}) > M(f_j(\mathbf{x}) - f_j(\mathbf{x}^*))$$

for all j such that $f_j(\mathbf{x}^*) < f_j(\mathbf{x})$. We can now write

$$f_i(\mathbf{x}^*) - f_i(\mathbf{x}) > (k-1)\frac{w_j}{w_i}(f_j(\mathbf{x}) - f_j(\mathbf{x}^*)).$$

After multiplying both sides by $w_i/(k-1) > 0$, we get

$$\frac{w_i}{k-1}(f_i(\mathbf{x}^*) - f_i(\mathbf{x})) > w_j(f_j(\mathbf{x}) - f_j(\mathbf{x}^*)) \ \Big(> 0 \geq w_l(f_l(\mathbf{x}) - f_l(\mathbf{x}^*))\Big),$$

where l differs from the fixed index i and the indices j, which were specified earlier. After this reasoning we can sum over all $j \neq i$ and obtain

$$w_i(f_i(\mathbf{x}^*) - f_i(\mathbf{x})) > \sum_{\substack{j=1\\ j\neq i}}^{k}(w_j(f_j(\mathbf{x}) - f_j(\mathbf{x}^*))),$$

which means

$$\sum_{j=1}^{k} w_j f_j(\mathbf{x}^*) > \sum_{j=1}^{k} w_j f_j(\mathbf{x}).$$

Here we have a contradiction to the assumption that $\mathbf{x}^*$ is a solution of the weighting problem. Thus, $\mathbf{x}^*$ has to be properly Pareto optimal. □

Theorem 3.1.7. If the multiobjective optimization problem is convex, then the condition in Theorem 3.1.6 is also necessary.

Proof. See Geoffrion (1968) or Chou et al. (1985).

Corollary 3.1.8. A necessary and sufficient condition for a point to be a properly Pareto optimal solution of an MOLP problem is that it is a solution of a weighting problem with all the weighting coefficients being positive.

The ratio of the weighting coefficients gives an upper bound to global trade-offs.

Theorem 3.1.9. Let $\mathbf{x}^*$ be a solution of weighting problem (3.1.1), when all the weighting coefficients w_i, $i = 1, \ldots, k$, are positive. Then

$$\Lambda_{ij}^{G}(\mathbf{x}^*) \leq \max_{\substack{i=1,\ldots,k\\ i\neq j}} \frac{w_j}{w_i}$$

for every $i, j = 1, \ldots, k$, $i \neq j$.

Proof. See Kaliszewski (1994, p. 93).

Some results concerning weak, proper and Pareto optimality of the solutions obtained by the weighting method are combined in Wierzbicki (1986b). Proper Pareto optimality and the weighting method are also discussed in Belkeziz and Pirlot (1991) and Luc (1995). The weighting method is used in Isermann (1974) in proving that for linear multiobjective optimization problems all the Pareto optimal solutions are also properly Pareto optimal. More results concerning MOLP problems and the weighting method are assembled in Chankong and Haimes (1983a, b, pp. 153–159).

3.1.2. Applications and Extensions

As far as applications are concerned, the weighting method is used to generate Pareto optimal solutions in Sadek et al. (1988–89) in solving a problem of the optimal control of a damped beam. The weighting method is also applied in Weck and Förtsch (1988) to structural systems in the optimization of a spindle bearing system and in the optimization of a table, as well as in ReVelle (1988), where reductions in strategic nuclear weapons for the two superpowers are examined. Furthermore, Pareto optimal solutions are generated for an anti-lock brake system control problem by the weighting method in Athan and Papalambros (1997). However, no attention is paid to the possible nonconvexity of the problem. In addition, the weighting method is an essential component in the determination of the optimal size of a batch system in Friedman and Mehrez (1992) and a fuzzy optimal design problem concerning a bridge is solved in Ohkubo et al. (1998).

Linear problems with two objective functions are studied in Gass and Saaty (1955). The systematic generation of the Pareto optimal set is possible in these problems by parametric optimization. The Pareto optimal set of multiobjective optimization problems with convex quadratic objective functions and linear equality constraints is characterized analytically in Goh and Yang (1996). The characterization involves the weighting method and active set methods.

Systematic ways of perturbing the weights to obtain different Pareto optimal solutions are suggested in Chankong and Haimes (1983a, b, pp. 234–236). In addition, an algorithm for generating different weighting coefficients automatically for convex (nonlinear) problems to produce an approximation of the Pareto optimal set is proposed in Caballero et al. (1997).

A method for reducing the Pareto optimal set (of an MOLP problem) before it is presented to the decision maker is suggested in Soloveychik (1983). Pareto optimal solutions are first generated by the weighting method. Then, statistical analysis (factor analysis) is used to group and partition the Pareto optimal set into groups of relatively homogeneous elements. Finally, typical solutions from the groups are chosen and presented to the decision maker.

It is suggested in Koski and Silvennoinen (1987) that the weighting method can be used to reduce the number of the objective functions before the actual solution process. The original objective functions are divided into groups such that a linear combination of the objective functions in each group forms a new objective function, and these new objective functions form a new multiobjective optimization problem. It is stated that every Pareto optimal solution of the new problem is also a Pareto optimal solution of the original problem, but the reverse result is not generally true.

As mentioned earlier, the weighting vector that produces a certain Pareto optimal solution is not necessarily unique. This is particularly true for linear problems. A method is presented in Steuer (1986, pp. 183–187) for determining ranges for weighting vectors that produce the same solution. Note that some weighting vectors may produce unbounded single objective optimization problems. This does not mean that the problem may not have feasible solutions with some other weighting vectors.

A property related to producing different Pareto optimal solutions by altering the weighting coefficients is the weak stability of the system. On the one hand, a small change in the weighting coefficients may cause big changes in the objective vectors. On the other hand, dramatically different weighting coefficients may produce nearly similar objective vectors. The reason is that the weighting problem is not a Lipschitzian function of the weighting coefficients. In addition, it is emphasized and illustrated in Das and Dennis (1997) that an evenly distributed set of weighting vectors does not necessarily produce an evenly distributed representation of the Pareto optimal set, even if the problem is convex. Further, Das and Dennis demonstrate how an even spread of Pareto optimal solutions is obtained only for very special shapes of Pareto optimal sets. The treatment concerns two objective functions.

An entropy-based formulation of the weighting method is suggested in Sultan and Templeman (1996). The entropy-based objective function to be optimized has only one parameter no matter what the number of the original objective functions is. A representation of the Pareto optimal set can be generated by varying the value of the single parameter. The properties of the suggested method are the same as those of the weighting method, for example, all the Pareto optimal solutions of nonconvex problems cannot be found.

3.1.3. Weighting Method as an A Priori Method

The weighting method can be used so that the decision maker specifies a weighting vector representing her or his preference information. In this case, the weighting problem can be considered (a negative of) a value function (remember that value functions are maximized). Note that according to Remark 2.8.7 of Part I, the weighting coefficients provided by the decision maker are now nothing but marginal rates of substitution ($m_{ij} = w_j/w_i$). When the weighting method is used in this fashion, it can be considered to belong to the class of a priori methods. Related to this, a method for assisting in the determination of the weighting coefficients is presented in Batishchev et al. (1991). This method can also be extended into an interactive form by letting the decision maker modify the weighting vectors after each iteration.

The objective functions should be normalized or scaled so that their objective values are approximately of the same magnitude (see Subsection 2.4.3 in Part I). Only in this way can one control and manoeuvre the method to produce solutions of a desirable nature in proportion to the ranges of the obfunction (see Section 2.6 in Part I). This is in many cases a rather simplifying assumption. In addition, it must be noted that altering the weighting vectors linearly does not have to mean that the values of the objective functions also change linearly. It is, moreover, difficult to control the direction of the solutions by the weighting coefficients, as illustrated in Nakayama (1995).

3.2. ε-Constraint Method

In the ε-constraint method, introduced in Haimes et al. (1971), one of the objective functions is selected to be optimized and all the other objective functions are converted into constraints by setting an upper bound to each of them. The problem to be solved is now of the form

$$\begin{aligned} &\text{minimize} && f_\ell(\mathbf{x}) \\ (3.2.1) \qquad &\text{subject to} && f_j(\mathbf{x}) \le \varepsilon_j \ \text{ for all } \ j = 1, \dots, k, \ \ j \ne \ell, \\ &&& \mathbf{x} \in S, \end{aligned}$$

where $\ell \in \{1, \dots, k\}$. Problem (3.2.1) is called an *ε-constraint problem*.

An alternative formulation is proposed in Lin (1976a, b), where proper equality constraints are used instead of the above-mentioned inequalities. The solutions obtained by this proper equality method are Pareto optimal under certain assumptions. Here, however, we concentrate on formulation (3.2.1).

3.2.1. Theoretical Results on Weak and Pareto Optimality

We begin by proving a result concerning weak Pareto optimality.

Theorem 3.2.1. The solution of ε-constraint problem (3.2.1) is weakly Pareto optimal.

Proof. Let $\mathbf{x}^* \in S$ be a solution of the ε-constraint problem. Let us assume that $\mathbf{x}^*$ is not weakly Pareto optimal. In this case, there exists some other $\mathbf{x} \in S$ such that $f_i(\mathbf{x}) < f_i(\mathbf{x}^*)$ for all $i = 1, \dots, k$.

This means that $f_j(\mathbf{x}) < f_j(\mathbf{x}^*) \le \varepsilon_j$ for all $j = 1, \dots, k$, $j \ne \ell$. Thus $\mathbf{x}$ is feasible with respect to the ε-constraint problem. While in addition $f_\ell(\mathbf{x}) < f_\ell(\mathbf{x}^*)$, we have a contradiction to the assumption that $\mathbf{x}^*$ is a solution of the ε-constraint problem. Thus, $\mathbf{x}^*$ has to be weakly Pareto optimal. □

Next, we handle Pareto optimality and the ε-constraint method.

Theorem 3.2.2. A decision vector $\mathbf{x}^* \in S$ is Pareto optimal if and only if it is a solution of ε-constraint problem (3.2.1) for every $\ell = 1, \dots, k$, where $\varepsilon_j = f_j(\mathbf{x}^*)$ for $j = 1, \dots, k$, $j \ne \ell$.

shown by a bold line. The upper bound level ε^1 is too tight and so the feasible region is empty. On the other hand, the level ε^4 does not restrict the region at all. If it is used as the upper bound, the point $\mathbf{z}^4$ is obtained as a solution. It is Pareto optimal according to Theorem 3.2.4. Correspondingly, for the upper bound ε^3 the point $\mathbf{z}^3$ is obtained as a Pareto optimal solution. The point $\mathbf{z}^2$ is the optimal solution for the upper bound ε^2. Its Pareto optimality can be proved according to Theorem 3.2.3. Theorem 3.2.2 can be applied as well.

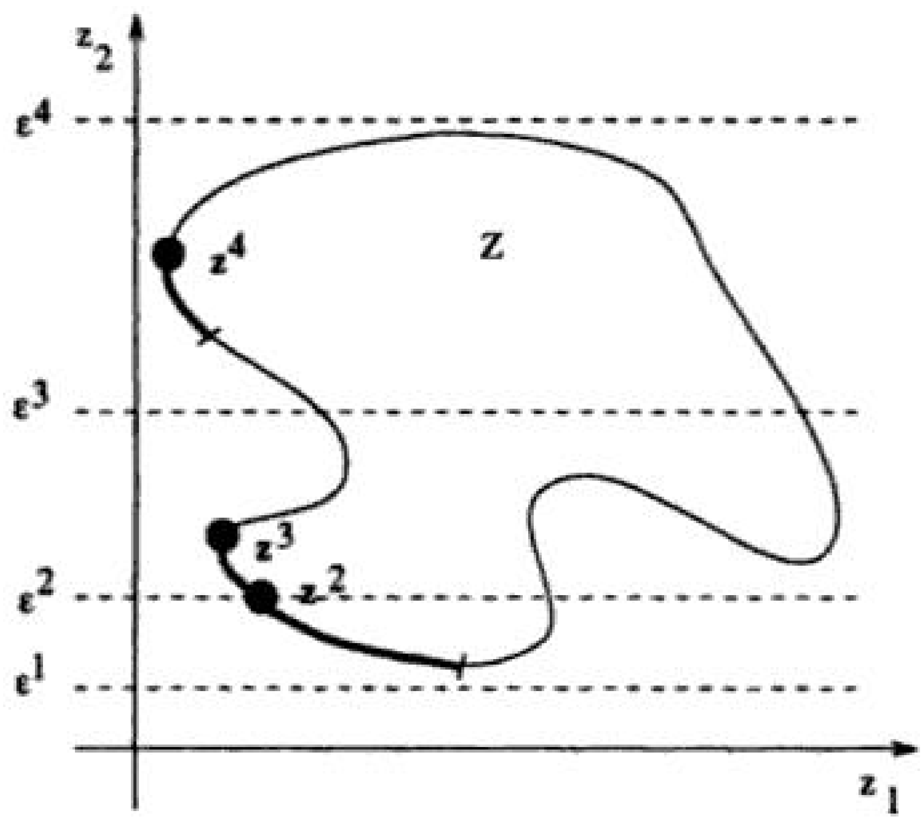

Figure 3.2.1. Different upper bounds for the ε-constraint method.

To ensure that a solution produced by the ε-constraint method is Pareto optimal, we have to either solve k different problems or obtain a unique solution. In general, uniqueness is not necessarily easy to verify. However, if for example, the problem is convex and the function f_ℓ to be minimized is strictly convex, we know that the solution is unique without further checking (see Chankong and Haimes (1983b, p. 131)).

According to Theorem 3.2.1 we know that the ε-constraint method produces weakly Pareto optimal solutions without any additional assumptions. We can show that any weakly Pareto optimal solution can be found with the ε-constraint method (for some objective function to be minimized) if the feasible region is convex and all the objective functions are quasiconvex and strictly quasiconvex. This result is derived in Ruíz-Canales and Rufián-Lizana (1995) and a shortened proof is also given in Luc and Schaible (1997). The value of this result is somewhat questionable because usually we are interested in Pareto optimal solutions and we know that any of them can be found with the ε-constraint method.

3.2.2. Connections with the Weighting Method

The relationships between the weighting method and the ε-constraint method are presented in the following theorems.

Theorem 3.2.5. Let $\mathbf{x}^* \in S$ be a solution of weighting problem (3.1.1) and $\mathbf{0} \leq \mathbf{w} \in \mathbf{R}^k$ be the corresponding weighting vector. Then

(1) if $w_\ell > 0$, $\mathbf{x}^*$ is a solution of the ε-constraint problem for f_ℓ as the objective function and $\varepsilon_j = f_j(\mathbf{x}^*)$ for $j = 1, \ldots, k$, $j \neq \ell$; or

(2) if $\mathbf{x}^*$ is a unique solution of weighting problem (3.1.1), then $\mathbf{x}^*$ is a solution of the ε-constraint problem when $\varepsilon_j = f_j(\mathbf{x}^*)$ for $j = 1, \ldots, k$, $j \neq \ell$ and for every f_ℓ, $\ell = 1, \ldots, k$, as the objective function.

Proof. Let $\mathbf{x}^* \in S$ be a solution of the weighting problem for some weighting vector $\mathbf{0} \leq \mathbf{w} \in \mathbf{R}^k$.

(1) We assume that $w_\ell > 0$. We have

$$\sum_{i=1}^{k} w_i f_i(\mathbf{x}) \geq \sum_{i=1}^{k} w_i f_i(\mathbf{x}^*) \tag{3.2.2}$$

for all $\mathbf{x} \in S$.

Let us suppose that $\mathbf{x}^*$ is not a solution of the ε-constraint problem. Then there exists a point $\hat{\mathbf{x}} \in S$ such that $f_\ell(\hat{\mathbf{x}}) < f_\ell(\mathbf{x}^*)$ and $f_j(\hat{\mathbf{x}}) \leq f_j(\mathbf{x}^*)$ when $j = 1, \ldots, k$, $j \neq \ell$. We assumed that $w_\ell > 0$ and $w_i \geq 0$ when $i \neq \ell$. Now we have

$$0 > w_\ell(f_\ell(\hat{\mathbf{x}}) - f_\ell(\mathbf{x}^*)) + \sum_{\substack{i=1 \\ i \neq \ell}}^{k} w_i(f_i(\hat{\mathbf{x}}) - f_i(\mathbf{x}^*)),$$

which is a contradiction with inequality (3.2.2). Thus $\mathbf{x}^*$ is a solution of the ε-constraint problem.

(2) If $\mathbf{x}^*$ is a unique solution of the weighting problem, then for all $\mathbf{x} \in S$

$$\sum_{i=1}^{k} w_i f_i(\mathbf{x}^*) < \sum_{i=1}^{k} w_i f_i(\mathbf{x}). \tag{3.2.3}$$

If there is some objective function f_ℓ such that $\mathbf{x}^*$ does not solve the ε-constraint problem when f_ℓ is to be minimized, then we can find a solution $\hat{\mathbf{x}} \in S$ such that $f_\ell(\hat{\mathbf{x}}) < f_\ell(\mathbf{x}^*)$ and $f_j(\hat{\mathbf{x}}) \leq f_j(\mathbf{x}^*)$ when $j \neq \ell$. This means that for any $\mathbf{w} \geq \mathbf{0}$ is $\sum_{i=1}^{k} w_i f_i(\hat{\mathbf{x}}) \leq \sum_{i=1}^{k} w_i f_i(\mathbf{x}^*)$. This contradicts inequality (3.2.3). Thus $\mathbf{x}^*$ is a solution of the ε-constraint problem for all f_ℓ to be minimized. □

The next theorem is valid for convex problems.

Theorem 3.2.6. Let the multiobjective optimization problem be convex. If $\mathbf{x}^* \in S$ is a solution of ε-constraint problem (3.2.1) for any given f_ℓ to be minimized and $\varepsilon_j = f_j(\mathbf{x}^*)$ for $j = 1, \dots, k$, $j \neq \ell$, then there exists a weighting vector $\mathbf{0} \leq \mathbf{w} \in \mathbf{R}^k$, $\sum_{i=1}^k w_i = 1$, such that $\mathbf{x}^*$ is also a solution of weighting problem (3.1.1).

Proof. The proof needs a so-called generalized Gordan theorem. See Chankong and Haimes (1983b, p. 121) and references therein.

We have now appropriate tools for proving Theorem 3.1.4 from the previous section.

Proof. (Proof of Theorem 3.1.4) Since $\mathbf{x}^*$ is Pareto optimal, it is by Theorem 3.2.2 a solution of the ε-constraint problem for every objective function f_ℓ to be minimized. The proof is completed with the aid of the convexity assumption and Theorem 3.2.6. □

A diagram representing several results concerning the characterization of Pareto optimal solutions and the optimality conditions of the weighting method, the ε-constraint method and a so-called jth Lagrangian method, their relations and connections is presented in Chankong and Haimes (1982, 1983b, pp. 119). The jth Lagrangian method, presented in Benson and Morin (1977), means solving the problem

$$\begin{aligned} &\text{minimize} && f_j(\mathbf{x}) + \sum_{\substack{i=1 \\ i \neq j}}^{k} u_i f_i(\mathbf{x}) \\ &\text{subject to} && \mathbf{x} \in S, \end{aligned} \tag{3.2.4}$$

where $\mathbf{u} = (u_1, \dots, u_{j-1}, u_{j+1}, \dots, u_k)^T$ and $u_i \geq 0$ for all $i \neq j$. The jth Lagrangian method is from a computational viewpoint almost equal to the weighting method. This is why it is not studied more closely here. Chankong and Haimes have treated the problems separately to emphasize two ways of arriving at the same point.

3.2.3. Theoretical Results on Proper Pareto Optimality

Let us now return to the ε-constraint problem and the proper Pareto optimality of its solutions. In Benson and Morin (1977), an auxiliary function, called the ***perturbation function***, $v\colon \mathbf{R}^{k-1} \to \mathbf{R}$ associated with the ε-constraint problem is defined in the form (modified here for the minimization problem)

$$v(\mathbf{y}) = \inf_{\mathbf{x} \in S} \{ f_\ell(\mathbf{x}) \mid f_j(\mathbf{x}) - \varepsilon_j \leq y_j \ \text{ for all } \ j = 1, \dots, k, \ j \neq \ell \}.$$

Naturally, the optimal value of the objective function of the ε-constraint problem is $v(\mathbf{0})$. We can now define the stability of ε-constraint problems.

NONLINEAR MULTIOBJECTIVE OPTIMIZATION

Kaisa M. Miettinen

Problems with multiple objectives and criteria are generally known as multiple criteria optimization or multiple criteria decision-making (MCDM) problems. These types of problems typically focus on linear programming methods. However, many phenomena are of a nonlinear nature, which is why we need tools for nonlinear programming capable of handling several conflicting or incommensurable objectives. In this case, methods of traditional single objective optimization are not enough; we need new ways of thinking, new concepts, and new methods—like nonlinear multiobjective optimization. NONLINEAR MULTIOBJECTIVE OPTIMIZATION provides an extensive, up-to-date, self-contained and consistent survey, review of the literature and of the state of the art on nonlinear (deterministic) multiobjective optimization, its methods, its theory and its background.

The amount of literature on multiobjective optimization is immense. The treatment in this book is based on approximately 1500 publications in English printed mainly after the year 1980. Problems related to real-life applications often contain irregularities and nonsmoothnesses. The treatment of nondifferentiable multiobjective optimization in the literature is rather rare. For this reason, we also include in this book material about the possibilities, background, theory and methods of nondifferentiable multiobjective optimization. Even though this book is restricted to deterministic nonlinear multiobjective methods, we would like to emphasize that promising new developments in multiobjective methods can be undertaken in researching problems that are stochastic and fuzzy in nature.

This book is intended for both researchers and students in the areas of (applied) mathematics, engineering, economics, operations research and management science; it is meant for both professionals and practitioners in many different fields of application. The intention has been to provide a consistent summary that may help in selecting an appropriate method for the problem to be solved. The extensive bibliography will be of value to researchers.

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